THE EFFECT OF INTERSTELLAR MATTER ON CLIMATIC VARIATION

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1. Introduction

There is direct astronomical evidence for the existence of diffuse clouds of matter in interstellar space. Any section of the Milky Way containing a large number of stars usually shows regions in which no stars appear, and the extent of these patches is often large compared with the average apparent distance between the stars themselves (see, for example, Russell, Dugan, and Stewart (2), p. 820). The existence of the so-called cosmical cloud in interstellar space, sharing in the general motion of the galaxy, is now well established, and observational investigation shows that the obscuration referred to above occurs also on a galactic scale. Thus the diffuse obscuring clouds appear as irregularities in the general cosmical cloud. The dimensions of such regions are comparable with the distances between the stars, and may be very much greater. In some instances the presence of such clouds is revealed by their illumination by a star or stars lying in, or near them, so that the matter then can be directly observed. In shape the clouds are very irregular; some appear like long dark lanes, while other tracts are devoid of any particular form.

Since the existence of such clouds appears to be general in the galaxy it is of importance to consider the effects that could be produced if a star passed through one of them. The frequency of such occurrences for a particular star would clearly depend upon the distribution of the clouds in space, and the intervals between these events would accordingly be irregular. But it is to be observed at once that the intervals would in general be of the order of the periods of time occurring in galactic problems, that is, of the order of 10^7 or 10^8 years, the average period of revolution of a star in the galaxy being about $2{\cdot}5 \times 10^8$ years.

The density of an obscuring cloud and the velocity that a star would have relative to it are known as far as orders of magnitude are concerned from astronomical considerations, and it is shown in the sequel that these clouds may have a considerable effect upon a star's radiation during the time of passage of the star through the cloud. The importance to terrestrial climate of such an effect upon the sun is at once evident, and it is to this aspect of the process that the present paper is directed, though it would seem that encounters between stars and the diffuse clouds may also have some bearing on questions of a more general astronomical nature. If any appreciable change in the sun's radiative power

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were brought about, it would certainly have considerable effects upon the climate of the earth, and the investigations contained in the present paper lead to the suggestion that the passage of the sun through nebulous clouds may give an account of certain epochs in the earth's history for which no tenable theory has hitherto been given. The actual mechanism that leads to the necessary increase in the solar output of energy is simply the addition to the sun of kinetic energy by the infall of part of the cloud; the gravitational energy is transformed into kinetic energy, and subsequently into radiant energy. Before proceeding with the details of this process we propose to give in brief outline some description of the climatic changes that have taken place during geological time.

2. PAST EPOCHS IN THE EARTH'S CLIMATIC HISTORY

(*a*) *The Ice Epochs*

At least eight Ice Epochs have been surmised from geological evidence (see, for example, Holmes (1), p. 217). These are separated by periods of the order of a hundred million years. More detailed evidence is available for the last of them, this being known as the Quaternary Ice Epoch, which seems to have comprised four separate ice ages and to have lasted as a whole not less than 200,000 years. Between these four ice ages the ice retreated to something like its present position, the intervening warm periods being long compared with the duration of the ice ages themselves.

(*b*) *The Carboniferous Epoch*

Much later than the last of the Pre-Cambrian Ice Epochs but before the Permian period, there occurred the Carboniferous period, during which the coal deposits were laid down. The coal originates from the plant life of the period, and judged by the thickness of the deposits this must have been prolific. From comparison with similar plant species in existence at the present time it is concluded that the climate during the Carboniferous Epoch must have been very warm and humid, these being the requirements now associated with such species. This is most remarkable when it is remembered that coal deposits are widespread over the earth and occur as far north as Spitzbergen, whose latitude is about 78° N. Since dynamical reasons make it certain that the land masses cannot have moved horizontally through appreciable distances (Jeffreys (3), p. 304), it follows that the climate over the whole earth must have been very much warmer than at the present time.

3. THEORIES OF CLIMATIC VARIATION

The theory that the hot climate of Carboniferous times was due to a period of great volcanic activity (which is known to have taken place about this time) cannot be described as indicating a satisfactory course of climatic variation, while

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the theories that have been put forward to explain the Ice Epochs also seem inadequate. The latter theories may be divided into two kinds: (1) terrestrial causes, and (2) astronomical causes.

It is not proposed to discuss the details of these theories apart from drawing attention once more to their defects that are overcome by the process arising from the present hypothesis.

The theories under (1) attribute variations in climate to differing terrestrial conditions at past times. Many of these, for example the change of sea-level, must have had some effect upon climate. It has been held that a small effect only is required to produce an ice age; the ice fields would begin to spread towards the equator, the process supposedly becoming cumulative, and if only small changes were required at the start, effects such as the change of sea-level might be sufficient. This theory would, however, meet difficulties in attempting to cope with the warm periods between the successive ice ages (of the Quaternary Ice Epoch) unless a suitable periodicity in the terrestrial conditions can be established.

The theories of the second type (2) based on extra-terrestrial causes, that have been proposed, have mainly invoked dynamical effects such as precession and changes in the solar eccentricity. But even if these suggestions were otherwise satisfactory, there is a fixed period associated with such dynamical motions, and this would be reflected in the effects of the climatic changes. The periods are not at all comparable with the irregular intervals at which exceptional climatic conditions have occurred as indicated by geographical and geological evidence.

4. Accretion of interstellar matter

The mechanism advocated in the present paper, and already adumbrated in the introduction, is of an astronomical nature and does not give rise to the defect of periodicity. It is shown that the effect of the passage of the sun through a cloud of diffuse matter is to increase the solar radiation, and that this process is able to bring about changes quantitatively in agreement with the requirements of climatic variation. The results obtained show that the percentage increase in the sun's radiation depends on the density of the cloud and on the velocity of the sun relative to it, being directly proportional to the first factor and inversely proportional to the cube of the latter factor. Thus slight changes in these factors bring about considerable ranges of variation in the solar radiation, and it may be caused to change from 0·1 to 1000% according to the density and velocity of the cloud. Meteorological considerations appear to indicate that an ice age is unlikely to result from small decreases in the amount of radiation received at the earth, for it is essential that evaporation should be increased in order to increase the precipitation of snow in those regions normally within the snow line, and the temperature of the oceans must accordingly be raised. On our view

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each Ice Epoch is to be regarded as due to the passage of the sun through a diffuse cloud with relative velocity sufficient to prevent more than a small percentage increase in the sun's output of radiation. Thus the difference of time between successive Ice Epochs will depend on the intervals between successive encounters of the sun with clouds of suitable relative velocity; from considerations of motion in the galaxy such periods would be expected to be of the order of 100 million years. (It is to be noticed that intervals as low as 10 million years are by no means excluded, for in its nature the process is far from periodic.) From their very existence it could be surmised that there will be considerable variations of density within any one cloud, and this is confirmed observationally. Any such large-scale differences in a cloud would have the effect of changing the sun's radiation by different amounts during its passage through the cloud. This would be capable of causing the type of climatic variations occurring in the Quaternary Ice Epoch. For, if an ice age is regarded as the period during which the sun's radiation changes slowly from its normal value to some suitable figure, a 10 % increase say, corresponding to the amount that would bring about a warm climate, then during the passage of the sun through a cloud in which the density increased towards the inner regions, two ice ages might occur separated by a warmer period. It would be necessary to assume only two such rises and falls in temperature (and density) in order to give the climatic form of the Quaternary Ice Epoch. (That a rise and fall in the solar radiation could produce such effects appears to have been first suggested by Simpson.) The two successive changes in the sun's total emission could then be accounted for by supposing the sun to have passed in turn through two adjacent clouds.

The Carboniferous Epoch would similarly be explicable as arising in the case when the sun passes through a cloud having either high average density or, what is more probable, low relative velocity, for then a large increase in the sun's radiation would occur. In order to produce a hot and humid climate even in polar regions the radiative power may have to be more than doubled. From the astronomical evidence it can be decided that vicissitudes of this intensity would be far less frequent, for the requisite density and velocity would be less likely to occur. For it turns out that the relative velocity of the sun and cloud required to produce an ice age is of just the order of magnitude of the solar velocity relative to the stars in its neighbourhood, while the value required to produce a period of exceedingly high terrestrial temperature is much smaller.

Consider now the process of accretion by the sun of the material of the cloud lying near its path. At first sight it might appear that the only particles of the cloud to be captured would be those which actually strike the sun as they attempt to describe orbits round its centre. The number of such particles under reasonable conditions would be quite insufficient to change the sun's radiation appreciably. It is clear, however, that some kind of condensation will be formed in the cloud

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by the sun's gravitational attraction, and the action of collisions in this condensation can be shown to give the sun an effective capture radius much larger than its ordinary radius.

(*a*) *Calculation of the capture radius of the sun*

Imagine the cloud to be streaming past the sun, from right to left in the figure, and let the velocity of any element of it relative to the sun when at great distances be v. Consider the part of the cloud that if undeflected by the sun would pass within a distance σ or less of its centre. It is clear that collisions will occur to the left of the sun because the attraction of the latter will produce two opposing streams of particles and the effect of such collisions is to destroy the angular

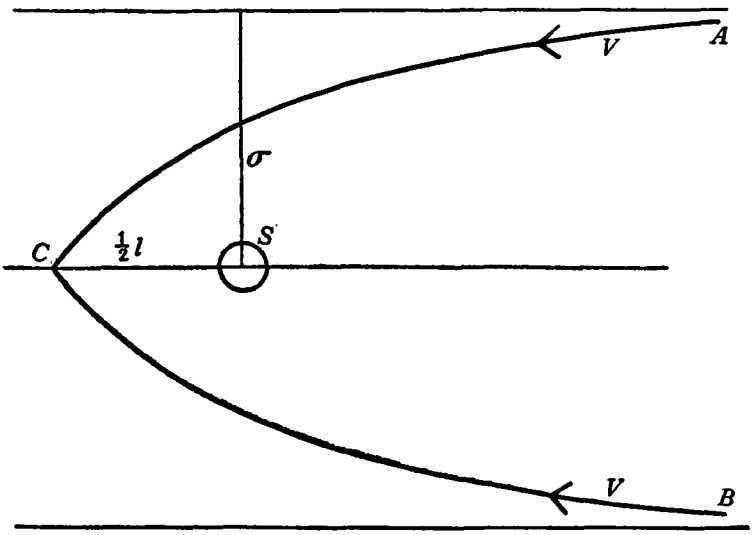


Fig. 1.

momentum of the particles about the sun. If after collision the surviving radial component of the velocity is insufficient to enable the particles to escape, such particles will eventually be swept into the sun. Suppose, for example, that an element of volume of the cloud A whose initial angular momentum is $v\sigma$ loses this momentum through its constituent particles suffering collisions at C; then the effective radius σ can be calculated such that the velocity radially at C is less than the escape velocity at this distance. The element describes a hyperbola whose equation, with the usual notation, may be written

$$\frac{l}{r} = 1 + e\cos\theta.$$

The direction parallel to the initial asymptote corresponds to

$$e\cos\theta + 1 = 0,$$

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and hence the direction SC corresponds to

$$e\cos\theta = 1, \quad \text{i.e. } e\sin\theta = (e^2-1)^{\frac{1}{2}}.$$

Thus the distance SC is simply $\frac{1}{2}l$. Since $r^2 d\theta/dt = h$, and is constant, the radial component of velocity is given by

$$\frac{dr}{dt} = \frac{eh}{l}\sin\theta. \qquad \checkmark$$

Now $h = \sqrt{(\mu l)} = v\sigma$, where μ/r^2 is the attraction of the sun at distance r. Hence, at C, dr/dt has the value v, the velocity of the cloud at infinity. The part of the cloud will accordingly fail to escape after collision if

$$v^2 < \frac{2\mu}{SC} = \frac{4\mu}{l} = \frac{4\mu^2}{v^2\sigma^2},$$

so that

$$\sigma < 2\mu/v^2.$$

Accordingly, the effective radius corresponds to the distance at which v is the parabolic velocity. If γ denotes the constant of gravitation and M is the mass of the sun then

$$\sigma = 2\gamma M/v^2.$$

For $v =$ 20 km. per sec., σ is as large as 1000 solar radii.

(*b*) *Collisions in the cloud*

What this foregoing calculation has shown is that, provided collisions occur with sufficient frequency to be effective in reducing the angular momentum of the particles about the sun, and if σ is defined as above, then nearly all particles passing within perpendicular distance σ when at great distances will eventually be captured by the sun. It is therefore necessary to consider the question of the number of collisions which a particle would undergo in describing an orbit round the sun. Since the cloud must be regarded as diffuse, the mean free path of a particle is long, but, on the other hand, the length of the path over which collisions take place is also very long. Consider the limiting case of a particle with initial angular momentum $v\sigma$. The length of path over which collisions are effective in reducing its angular momentum is of order σ. It is necessary only to show that the number of collisions is sufficiently great to justify the previous calculation of σ as the capture radius; in order to do this the problem may be conveniently simplified. Consider, therefore, a particle of velocity v fired into a tube of length σ and cross-section 1 cm.^{2}, through which the cloud is streaming with velocity v in the opposite direction. If the density of the cloud is known, the effective number of collisions is readily obtainable. If we assume that the density is 10^{-18} g. per cm.^{3} (Russell, Dugan and Stewart (2), p. 832), then if the cloud is composed of hydrogen the corresponding number N of atoms per cm.^{3} is about 6×10^5. (The assumption of atomic hydrogen is made in agreement with the supposition that the stars have evolved from primitive hydrogen nebulae.)

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Thus, in passing along the tube, the one hydrogen atom in which interest is fixed moves past a total number of particles at least as great as $2N\sigma$. If we take the radius of the hydrogen atom to be 10^{-8} cm., the number of collisions is accordingly $2\pi N\sigma \times 10^{-16}$. (A simple classical kinetic model is used for the collisions. If a quantum treatment is introduced the effective scattering of the hydrogen atom is greater than in the classical picture but is of the same order of magnitude.) If we express σ in terms of the mass of the sun and the relative velocity of the sun and cloud, the number of collisions is

$$4\pi\gamma NM/10^{16}v^2.$$

If we take

$$\gamma = \tfrac{20}{3} \times 10^{-8}, \quad N = 6 \times 10^5,$$

$$M = 2 \times 10^{33} \text{ g.} \quad \text{and} \quad v = 2 \times 10^6 \text{ cm. per sec.},$$

the number of collisions is about $2{\cdot}5 \times 10^4$. This number is certainly sufficiently high to destroy the angular momentum of such particles about the sun. Possibly the estimate of σ as the length of track over which collisions occur is too large, but even if it is reduced by a factor $\frac{1}{10}$ the number of collisions would nevertheless be of order 10^3. Particles that possess initial angular momentum less than $v\sigma$ have less length of effective track, but this is compensated by an increase in the density of the cloud towards the sun. Thus it may be concluded that, for a cloud of the assumed mean density, nearly all particles crossing a circle of radius $\sigma = 2\gamma M/v^2$ are captured by the sun. Since the number of collisions has been shown to be large, this conclusion must also be valid for considerable ranges in the assumed density and is probably applicable for densities as low as 10^{-23} g. per cm.^{3}.

The cloud has been regarded as un-ionized in calculating the collisions. For an ionized cloud the chief process in the present connexion will be the scattering of protons by protons, and it may be verified that this would be rather more effective in destroying the angular momentum of the particles of the cloud than collisions of neutral atoms.

(c) *The accretion of mass*

The important question is how to calculate the amount of kinetic energy that the material of the cloud captured by the sun will bring to the latter due to its fall on to the solar surface. Since the velocity of escape at the surface of the sun is so large, we can suppose that all the particles reaching the surface of the sun arrive with the escape velocity, that is, $6{\cdot}2 \times 10^7$ cm. per sec.; this is permissible provided that there is no other way in which the particle can get rid of its energy. For a hydrogen atom, having mass $1{\cdot}66 \times 10^{-24}$ g., the corresponding kinetic energy is slightly in excess of 3×10^{-9} ergs. Now the ionization energy of the hydrogen atom is about 4×10^{-11} ergs, hence it can be concluded that the particle cannot get rid of any appreciable portion of its energy by ionization processes

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before reaching the sun. Thus the obscuring effect of the particles lying between the earth and the sun would be quite negligible compared with the increase of the sun's radiation due to the kinetic energy of the particles. The sun must re-emit this energy.

The total number of particles reaching the sun per second is simply the number crossing an area bounded by a circle of radius σ perpendicular to the direction of v, the velocity of the cloud. Their mass is accordingly $\pi\sigma^2\rho v$, and, if we express σ in terms of the mass of the sun, then the mass added per second is $4\pi\gamma^2\rho M^2/v^3$, and this formula is true for all ρ, M, and v, so long as collisions remain important at distances comparable with σ. The kinetic energy brought to the sun per second is obtained at once, since we know the velocity of escape at the sun's surface. The energy is found to be

$$4 \times 10^{68}\rho v^{-3} \text{ ergs.}$$

For climatic effects to be appreciable, this amount of energy must be comparable with the sun's ordinary output. We see that the increase is more sensitive to changes in v than in ρ. Suppose then that $\rho = 10^{-18}$, for example, and consider the value which v would have to take in order to bring about a 10 % change in the sun's normal radiation, which means a change of about 4×10^{32} ergs per sec. Then

$$4 \times 10^{50} \times v^{-3} = 4 \times 10^{32},$$

and $v = 10$ km. per sec. For a 1 % increase, v would be approximately 20 km. per sec., while for a 1000 % increase v would be about 2 km. per sec. On the other hand, if $\rho = 10^{-20}$, an increase of 1 % would occur for a velocity of about 5 km. per sec.

Stellar considerations indicate that a velocity of 2×10^6 cm. per sec. would be about the most probable value for an encounter, and the agreement of the results of the present investigation as regards orders of magnitude is manifest. The velocity required to cause a tremendous change in the solar radiation is not unacceptably low, and the value obtained shows the degree of improbability that such changes would occur frequently. In conjunction with this remark it should be remembered that the tendency of the cloud would be to increase the velocity of approach of the sun; this is now considered.

(*d*) *Acceleration of the sun*

It is necessary to consider the interval for which the sun is likely to be exposed to an encounter. In general the time could be made of any desired duration by adjusting the size of the supposed cloud. But a massive cloud would increase the velocity of the sun relative to it, unless the sun was permanently inside it, and it appears that if the mass of the cloud were very much greater than that of the sun, the relative velocity might be increased by its attraction to a value much greater than the estimate of 10^6 cm. per sec. used above. This in turn would bring about only a very slight increase in the radiation of the sun; for a relative

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velocity of 10^7 cm. per sec. the increase is about 0·01 % only. Thus for an encounter to cause a suitable change in radiation, the alteration of the velocity of the sun must not exceed 10^6 cm. per sec. The effective mass of the cloud for this purpose will depend largely on its shape, and upon what region of it the sun happens to traverse, and may vary widely from its actual total mass.

Let m be of the order of the effective mass of the cloud, and let $2a$ be the length of the path which the sun travels in it. Then the gravitational potential at the surface is of the order of $\gamma m/a$, where γ is the usual constant of gravitation. If v is the velocity of the sun, the change Δv produced by the attraction of the cloud is therefore given, so far as order of magnitude is concerned, by

$$v\Delta v = \gamma m/a,$$

provided that Δv is not very much greater than v. If $v = 10^6$ cm. per sec., in accordance with the previous requirements, and if Δv is taken as 10^5 cm. per sec., then it is found that

$$m/a = 1{\cdot}5 \times 10^{18}.$$

Now m is of order $4\rho a^3$, and if we put $\rho = 10^{-18}$ we find that

$$a = 6 \times 10^{17} \text{ cm.}$$

The time taken to travel through the cloud is therefore of order $2a/v$, i.e. about 10^{12} sec. (about 10^5 years). From the way this has been computed, we see that this gives an estimate of the minimum time required in general to cross a cloud. Furthermore, if the cloud were patchy, an increase in the time of traversing it would also arise. The effective mass of the cloud is of order $4\rho a^3$ and this is about 500 times the mass of the sun. In view of the rough methods which the nature of the problem compels us to adopt, the agreement with the requirements of observation are completely in accordance with the proposed hypothesis.

(*e*) *Total mass accreted by the sun and the earth*

It is important to show also that the matter added to the sun during the whole period of its passage through the cloud will not appreciably affect its mass.

To change the solar radiation by 1 % requires an energy addition of $3{\cdot}78 \times 10^{31}$ ergs per sec., and, since the velocity at the surface of the sun is about 620 km. per sec., this amount implies an increase in mass of 2×10^{16} g. per sec. Hence in 10^{12} sec. the change is 2×10^{28} g., about 3 times the mass of the earth, i.e. about 10^{-5} times that of the sun itself. According to the mass-luminosity relation, the usual radiation (when the sun is not passing through a cloud) can be taken as being roughly proportional to a small power of the sun's mass, say $L \propto M^3$, and hence, even if considerable allowance is made for the material added being hydrogen, the change in luminosity could not be as great as 0·01 %. Thus the sun would revert to its former luminosity to within this margin.

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It is of some interest also to consider the accompanying change of mass of the earth. The velocity of the latter relative to the cloud would vary in any case, but at the earth's distance the cloud captured by the sun would presumably be mainly moving radially inwards with the velocity of escape at the earth's distance, which is over 40 km. per sec. But even if this effect is ignored and if the earth's velocity is taken as 2×10^6 cm. per sec. relative to the cloud, the addition of mass turns out to be insensibly small. For it has been seen that the amount added is proportional to the square of the mass of the central body, so that other things being equal the amount added by the sun will be 10^{11} times the amount added by the earth. Thus the earth would receive 2×10^{17} g. in 10^{12} sec., and this is less than 10^{-4} the mass of the terrestrial atmosphere. If allowance is made for the orbital motion of the earth and the radial motion of the cloud towards the sun, this would be reduced by a factor of about $\frac{1}{30}$ and the amount would be about a millionth the mass of the present atmosphere. This would represent a wholly negligible change, and since in addition it would take 10^5 years to be added, could itself produce no sensible climatic effects.

Accompanying the change in mass of the sun there will be a change in the mean distance of the earth. For a nearly circular orbit the change Δa of the major semi-axis a is given by

$$\frac{\Delta a}{a} = -\frac{\Delta M}{M}.$$

Hence the total change in a will be a decrease of the order of $10^{-5}a$. The consequent change in the solar radiation received at the earth will be of the same order, and is therefore quite inappreciable. It appears, therefore, that in the course of describing its orbit in the galaxy, the sun would change its luminosity whenever it happened to pass through a nebulous cloud of any appreciable density. The density of the cosmical cloud spread throughout the galaxy is of the order of 10^{-23} g. per c.c., and its presence will not affect the luminosity of the sun, but the existence of huge irregular tracts of much higher density is well known. For instance, the Orion nebula may have an average density as great as 10^{-18} g. per c.c. (Russell, Dugan and Stewart (2), p. 832). Such clouds are prevalent in the galaxy and it may reasonably be expected that the sun has undergone several encounters with clouds of these densities, since the period of geological time and the period of motion of the sun in the galaxy are comparable.

If the conclusions of this paper are correct the solar system must have emerged from such a cloud within the last few thousand years, since an ice age took place as recently as this. Thus the cloud responsible could not be as remote as the brightest stars. But, even if the density were as high as 10^{-18} g. per c.c., starlight in traversing its breadth, which must be of order 6×10^{17} cm., would have to pass through the equivalent of a layer of 1 cm. of gas of density 1 g. per c.c. Only one light quantum in a hundred would be absorbed in such a layer as far as visual

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frequencies are concerned, though higher energy quanta would be more strongly absorbed. Thus there would be no reason to suspect appreciable obscuration of the background stars to be caused by such a cloud.

We submit, therefore, that the present paper represents a discussion of a process that is of importance in considering the past history of the sun and that some, if not all, of the climatic variations occurring on earth may have resulted from the consequent changes in solar radiation.

5. Conclusion

The foregoing investigation relates to a process that seems hitherto to have been overlooked. The number of mechanisms that have been put forward as responsible for the Ice Epochs is manifold, but the present suggestion would seem to have the merit of giving effects quantitatively in agreement with the requirements and free from any periodicity. The present hypothesis is not advocated with a view to contradicting the opinions and results of those investigators who have confined their researches to terrestrial sources of climatic variation. The claim made is simply that an important process is brought to light and must be accorded some place in discussions of terrestrial climate.

6. Summary

The effect of interstellar matter on the sun's radiation is considered with a view to explaining changes in terrestrial climate. It appears that a star in passing through a nebulous cloud will capture an amount of material which by the energy of its fall to the solar surface can bring about considerable changes in the quantity of radiation emitted. The quantity of matter gathered in by the star depends directly on the density of the cloud and inversely on the cube of its velocity relative to the cloud. Thus vastly different effects on the solar radiation can be brought about under fairly narrow ranges of density and relative velocity (ranges that are in accordance with astronomical evidence). In this way the process is able to explain the small changes in the solar radiation that are necessary to produce an ice age and, under conditions less likely to have taken place frequently, the high increase in radiation required for the Carboniferous Epoch. Despite the large effects that the mechanism can bring about, it is shown that the mass of the sun does not undergo appreciable change and hence reverts to its former luminosity once the cloud has been traversed.

References

(1) Holmes, A. *The age of the Earth.* Nelson Classics. (1937.)
(2) Russell, H. N., Dugan, R. S. and Stewart, J. Q. *Astronomy.* Ginn and Company (1938).
(3) Jeffreys, H. *The Earth.* Cambridge (1929).

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